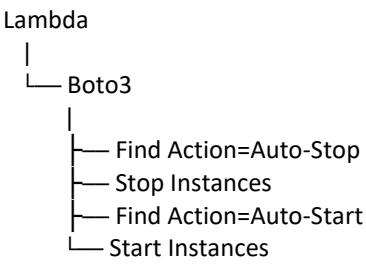
Assignment 1: Automated EC2 Start/Stop Using Lambda

Architecture



Step 1: Create EC2 Instances

Create two EC2 instances:

Instance 1

Tag:

Key: Action
Value: Auto-Stop

Instance 2

Tag:

Key: Action
Value: Auto-Start

Step 2: Create IAM Role

IAM → Roles → Create Role

Trusted Entity:

AWS Service
Lambda

Attach Policy:

AmazonEC2FullAccess

Role Name:

Lambda-EC2-Manager

Step 3: Create Lambda Function

Runtime:

Python 3.12

Execution Role:

Lambda-EC2-Manager

Step 4: Lambda Code

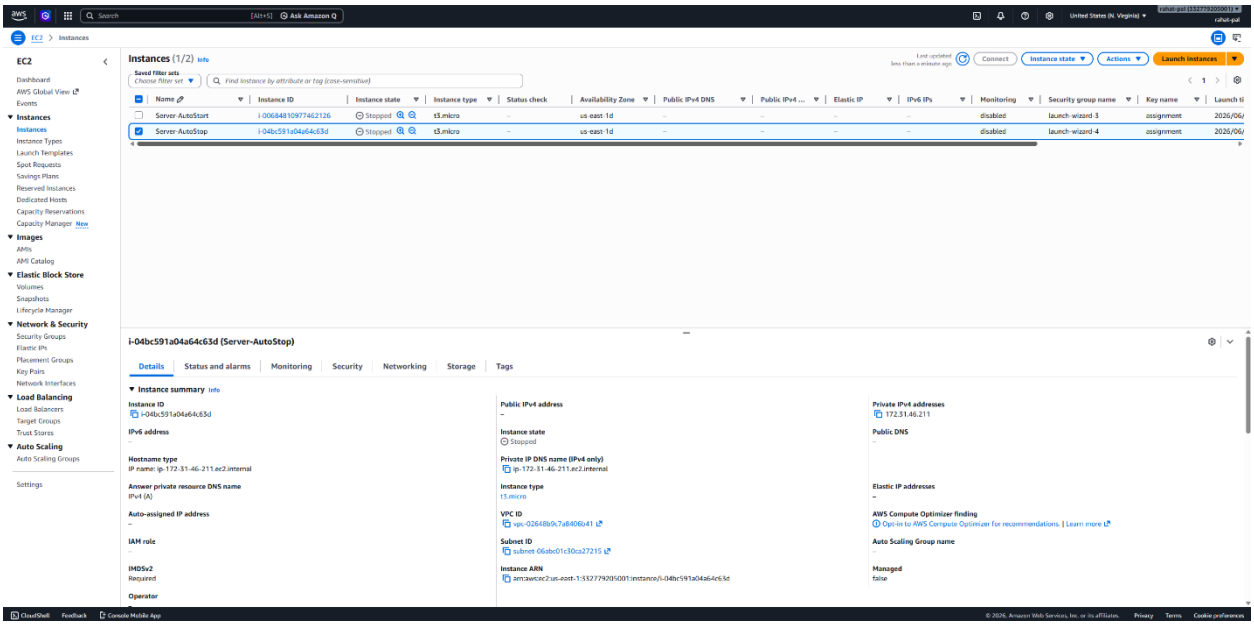
```
import boto3

ec2 = boto3.client('ec2')

def lambda_handler(event, context):

    stop_instances = []
    start_instances = []

    response = ec2.describe_instances(
        Filters=[
            {
                'Name': 'tag:Action',
                'Values': ['Auto-Stop']
            }
        ]
    )

    for reservation in response['Reservations']:
        for instance in reservation['Instances']:
            stop_instances.append(instance['InstanceId'])

    if stop_instances:
        ec2.stop_instances(InstanceIds=stop_instances)

    response = ec2.describe_instances(
```

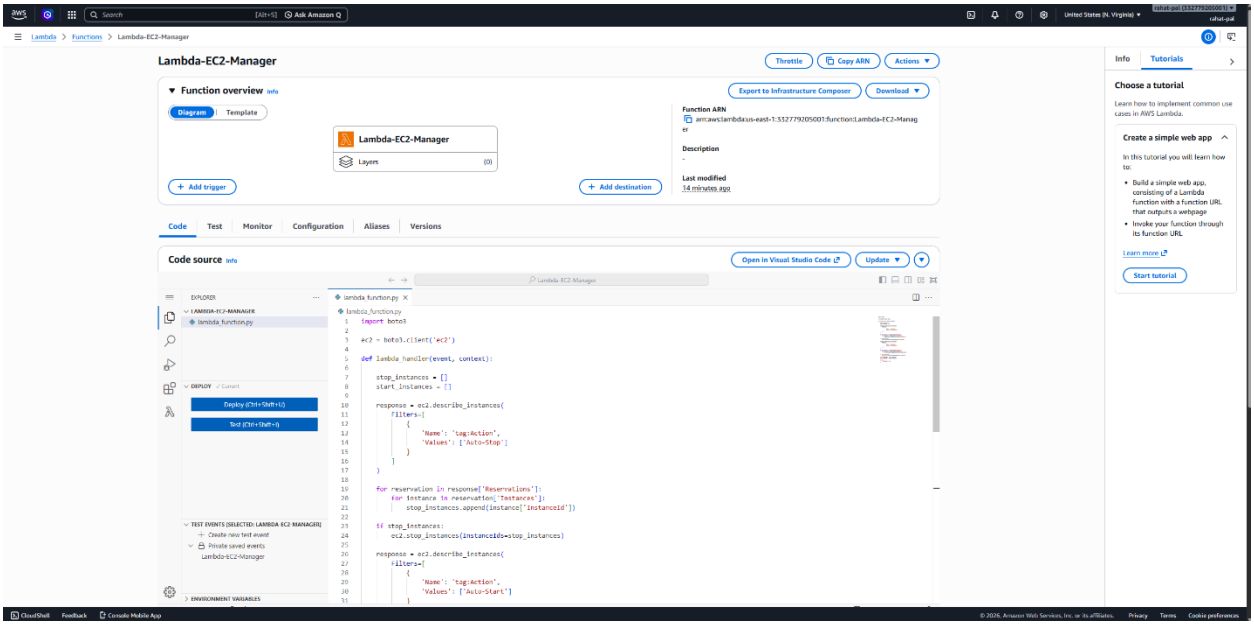
```
    Filters=[
        {
            'Name': 'tag:Action',
            'Values': ['Auto-Start']
        }
    ]
)

for reservation in response['Reservations']:
    for instance in reservation['Instances']:
        start_instances.append(instance['InstanceId'])

if start_instances:
    ec2.start_instances(InstanceIds=start_instances)

print("Stopped:", stop_instances)
print("Started:", start_instances)

return {
    'statusCode': 200
}
```

Expected Lambda Output

Stopped: ['i-0ab123456789']
Started: ['i-0cd987654321']

Screenshots




Lambda-EC2-Manager
Throttle
Copy ARN
Actions
Function overview
Export to Infrastructure Composer
Download
Diagram
Template
Lambda-EC2-Manager
Layers
Add trigger
Add destination
Function ARN
Description
Last modified
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Update
Info
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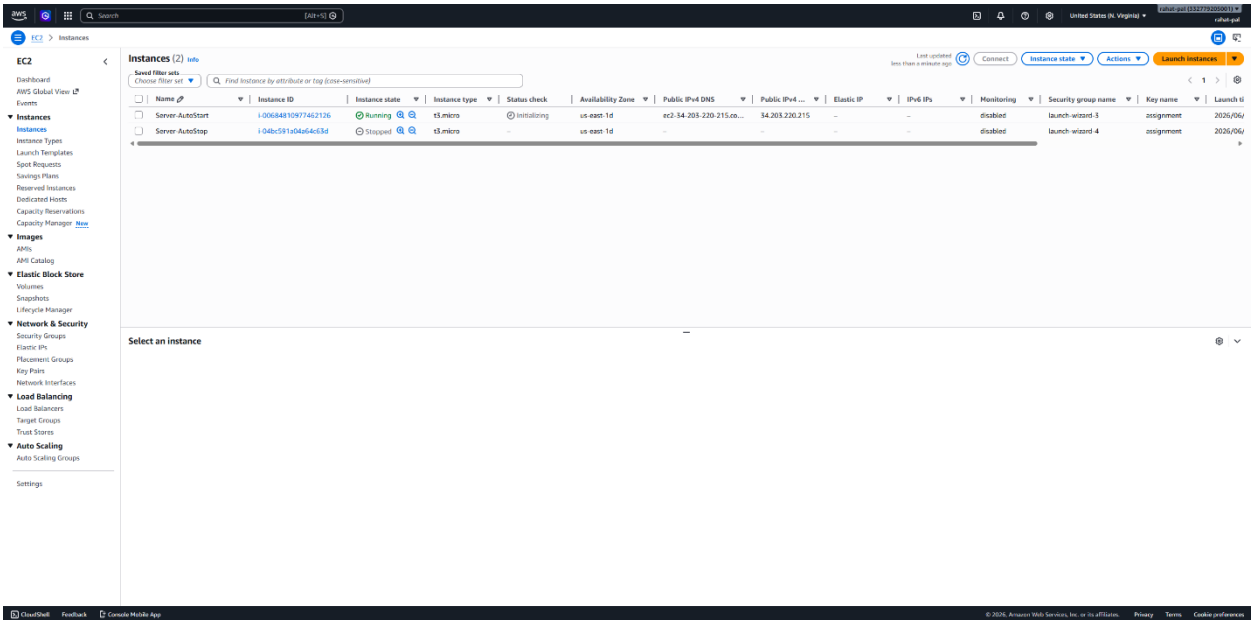

EC2
Instances
Launch instances
Server-AutoStart
Server-AutoStop
Select an instance

Assignment 2: S3 Cleanup (Delete Files Older Than 30 Days)

Step 1: Create Bucket

Example:

aws-cleanup-demo-bucket

Upload:

file1.txt
file2.txt
file3.txt

Step 2: IAM Role

Attach:

AmazonS3FullAccess

Step 3: Lambda Code

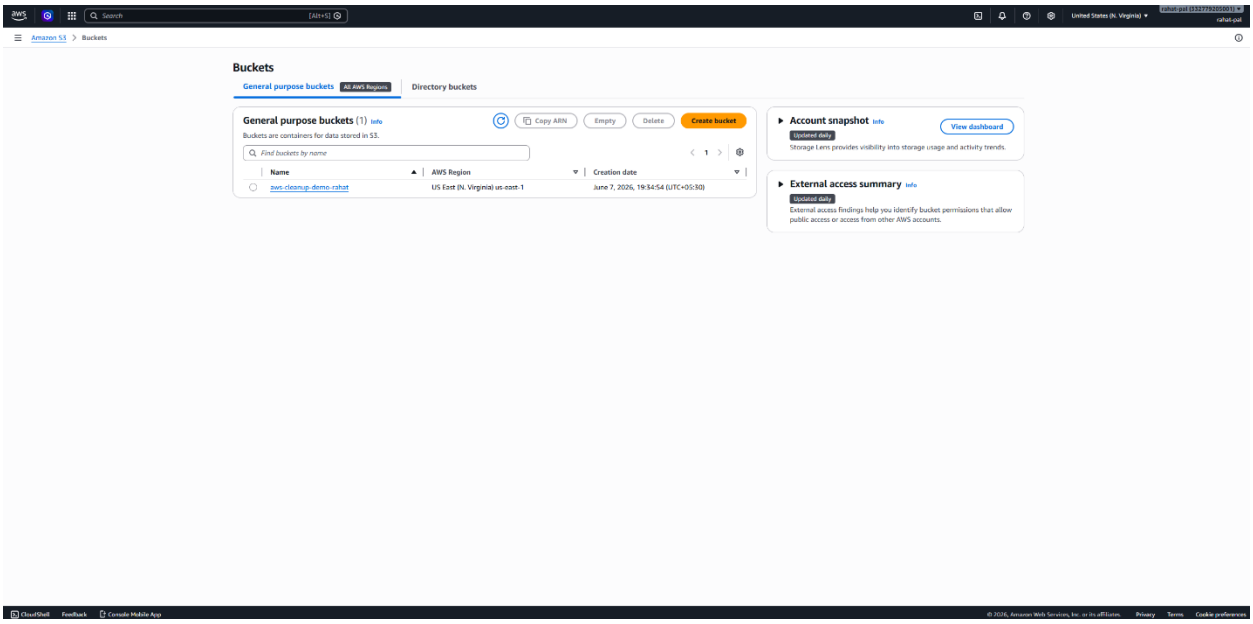
```
import boto3
from datetime import datetime, timezone, timedelta

s3 = boto3.client('s3')

BUCKET_NAME = "aws-cleanup-demo-bucket"

def lambda_handler(event, context):

    cutoff = datetime.now(timezone.utc) - timedelta(days=30)

    response = s3.list_objects_v2(Bucket=BUCKET_NAME)

    for obj in response.get('Contents', []):

        if obj['LastModified'] < cutoff:

            s3.delete_object(
                Bucket=BUCKET_NAME,
                Key=obj['Key']
```

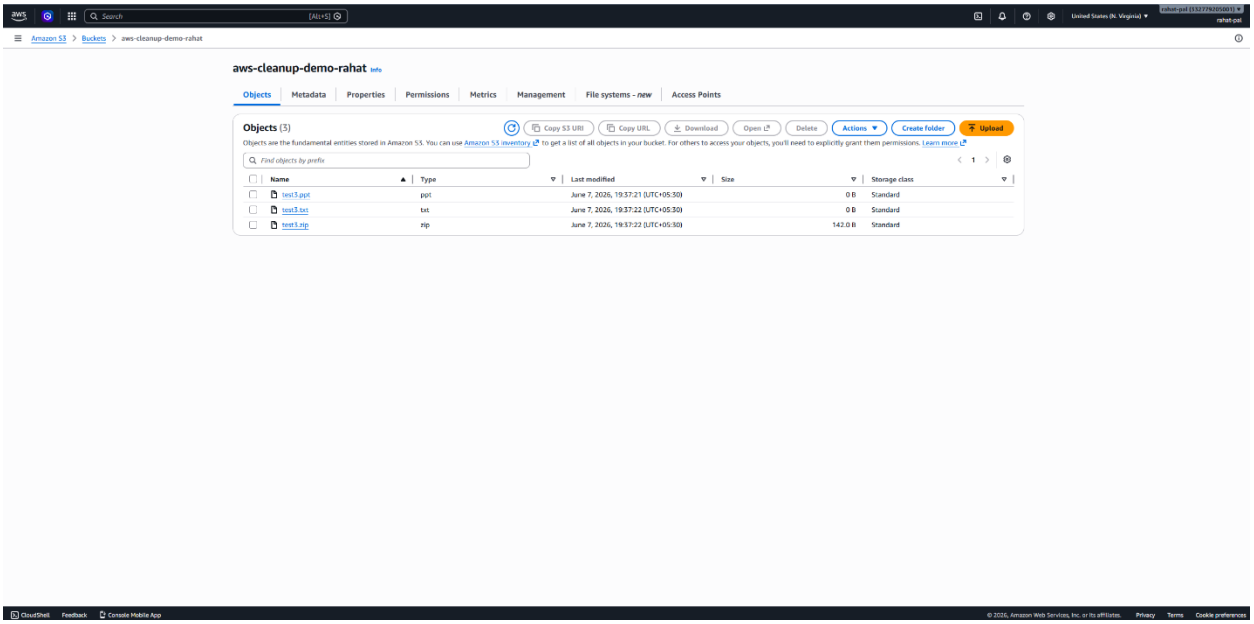
```
        )

        print(f"Deleted: {obj['Key']}")

    return {
        'statusCode': 200
    }
```

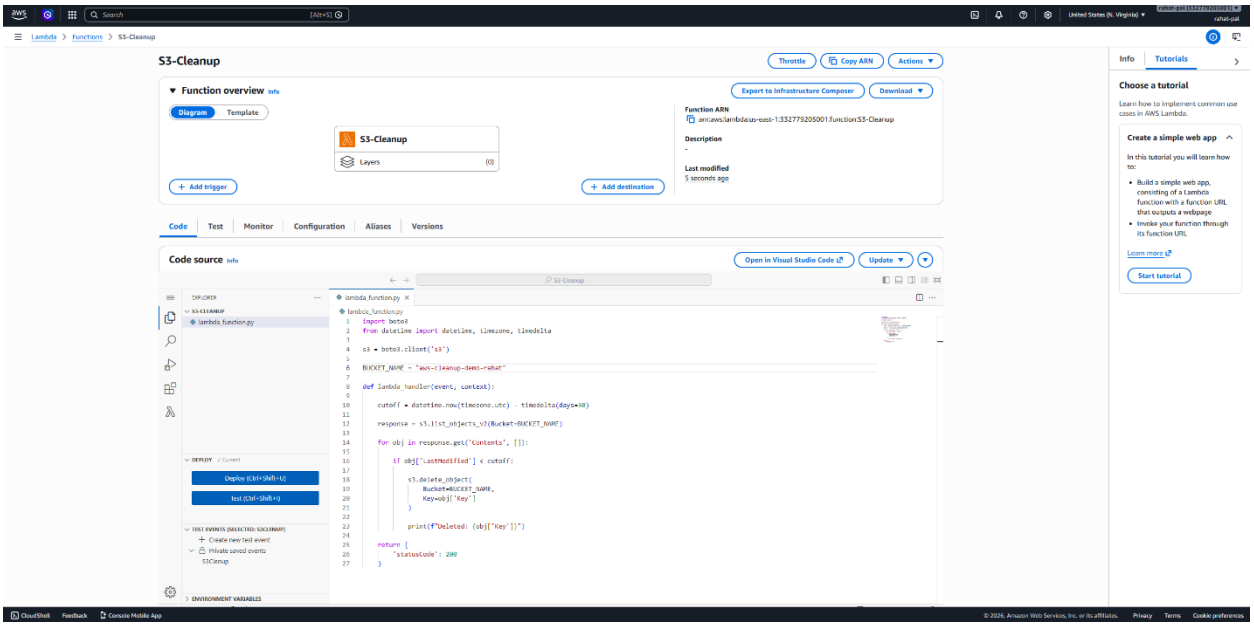
Expected Output

Deleted: old_backup.zip
Deleted: old_logs.txt

Screenshots




aws-cleanup-demo-rahat
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (3)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Name
Type
Last modified
Size
Storage class
test3.ppt
ppt
June 7, 2026, 19:37:21 (UTC+05:30)
0 B
Standard
test3.txt
txt
June 7, 2026, 19:37:22 (UTC+05:30)
0 B
Standard
test3.zip
zip
June 7, 2026, 19:37:22 (UTC+05:30)
142.0 B
Standard


S3-Cleanup
Throttle
Copy ARN
Actions
Function overview
Export to Infrastructure Composer
Download
Diagram
Template
S3-Cleanup
Layers
Add trigger
Add destination
Function ARN
arn:aws:lambda:us-east-1:332779205001:function:S3-Cleanup
Description
Last modified
5 seconds ago
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Update
lambda_function.py
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Info
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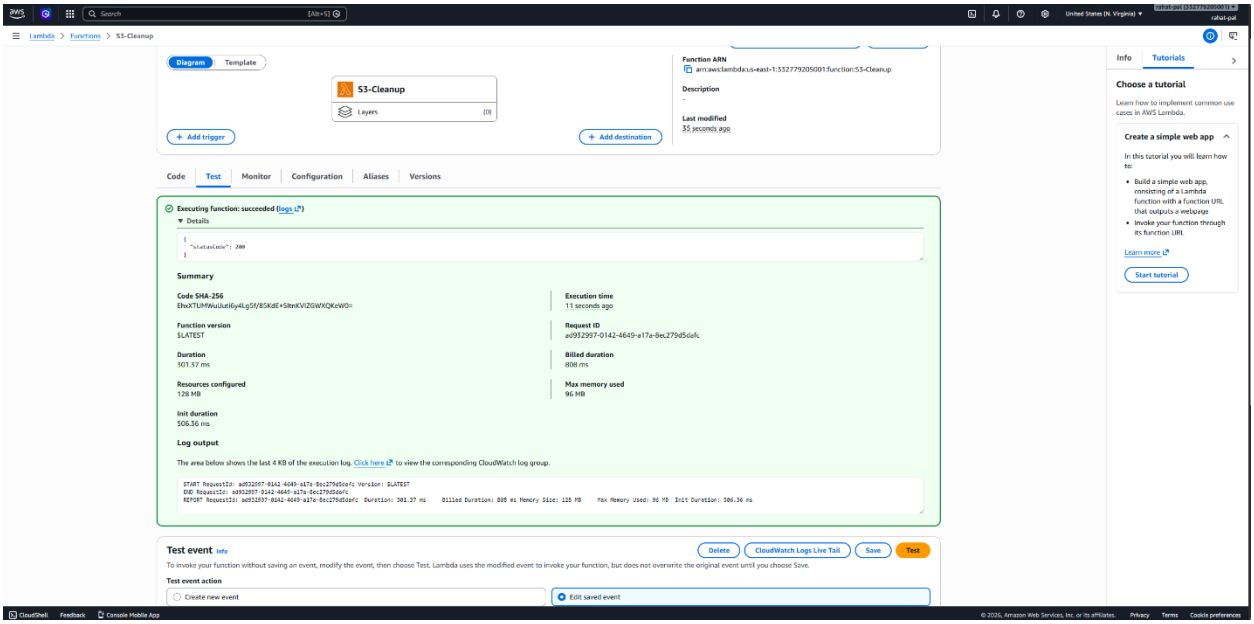
Lambda > Functions > S3-Cleanup
Diagram
Template
S3-Cleanup
Layers
(0)
+ Add trigger
+ Add destination
Function ARN
arn:aws:lambda:us-east-1:332779205001:function:S3-Cleanup
Description
-
Last modified
35 seconds ago
Info
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Invoke your function through its function URL
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Code
Test
Monitor
Configuration
Aliases
Versions
Executing function: succeeded (logs)
Details
{
"statusCode": 200
}
Summary
Code SHA-256
EhxXTUMWuUuuti6y4Lg5f/85KdE+SltnKVIZGWXQKcWO=
Execution time
11 seconds ago
Function version
$LATEST
Request ID
ad932997-0142-4649-a17a-8ec279d5dafc
Duration
301.57 ms
Billed duration
808 ms
Resources configured
128 MB
Max memory used
96 MB
Init duration
506.36 ms
Log output
The area below shows the last 4 KB of the execution log. Click here to view the corresponding CloudWatch log group.
START RequestId: ad932997-0142-4649-a17a-8ec279d5dafc Version: $LATEST
END RequestId: ad932997-0142-4649-a17a-8ec279d5dafc
REPORT RequestId: ad932997-0142-4649-a17a-8ec279d5dafc Duration: 301.57 ms Billed Duration: 808 ms Memory Size: 128 MB Max Memory Used: 96 MB Init Duration: 506.36 ms
Test event
Info
Delete
CloudWatch Logs Live Tail
Save
Test
To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.
Test event action
Create new event
Edit saved event
CloudShell
Feedback
Console Mobile App
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Assignment 3: Detect Unencrypted S3 Buckets

IAM Role

Attach:

AmazonS3ReadOnlyAccess

Lambda Code

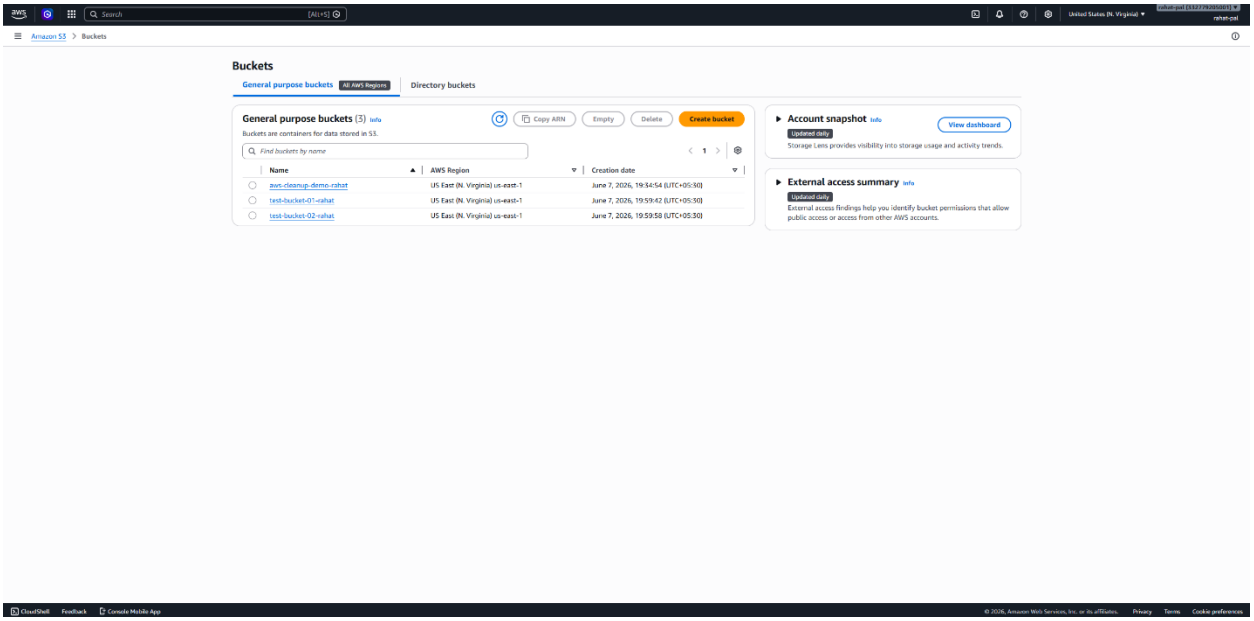
```
import boto3

from botocore.exceptions import ClientError


s3 = boto3.client('s3')


def lambda_handler(event, context):


    encrypted_buckets = []

    unencrypted_buckets = []


    buckets = s3.list_buckets()


    for bucket in buckets['Buckets']:

        bucket_name = bucket['Name']


        try:

            s3.get_bucket_encryption(Bucket=bucket_name)

            encrypted_buckets.append(bucket_name)
```

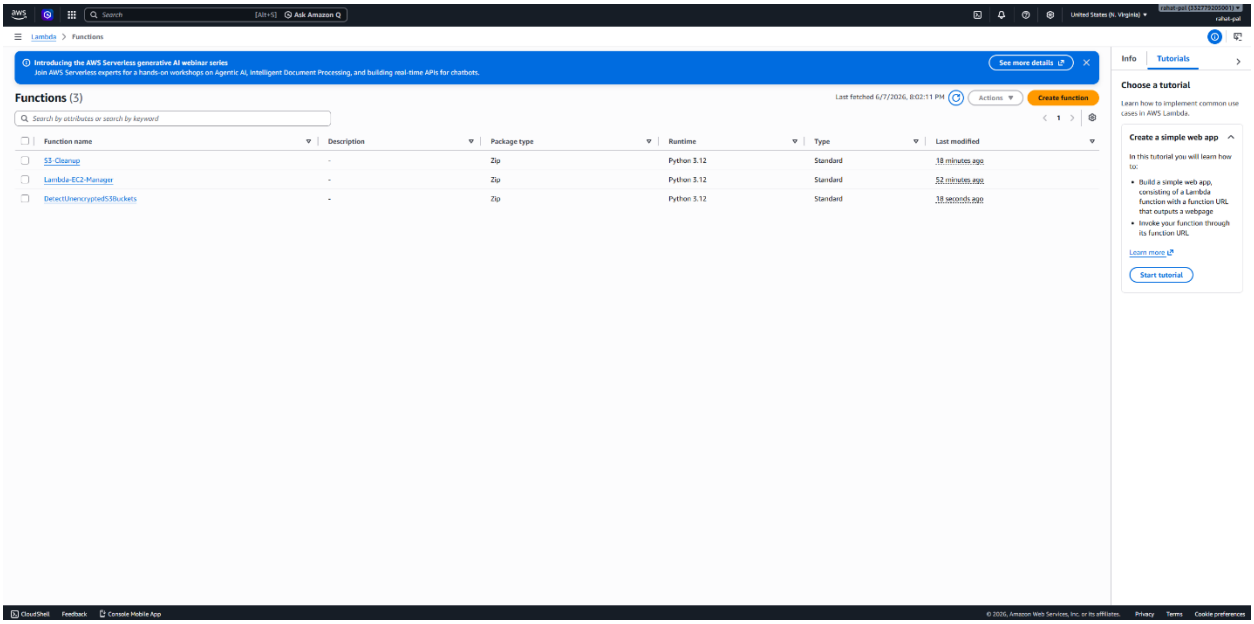
```
    except ClientError as e:
        error_code = e.response['Error']['Code']

        if error_code == 'ServerSideEncryptionConfigurationNotFoundError':
            unencrypted_buckets.append(bucket_name)

print("=== Encryption Report ===")

if unencrypted_buckets:
    print("\nUnencrypted Buckets:")
    for bucket in unencrypted_buckets:
        print(bucket)
else:
    print("\nNo unencrypted buckets found.")

print("\nEncrypted Buckets:")
for bucket in encrypted_buckets:
    print(f"{bucket} - Encrypted")

return {
    "statusCode": 200,
    "message": "All buckets checked successfully",
    "unencrypted_buckets": unencrypted_buckets,
```

```
        "encrypted_buckets": encrypted_buckets
    }
```

Expected Output:

```
{
  "statusCode": 200,
  "message": "All buckets checked successfully",
  "unencrypted_buckets": [],
  "encrypted_buckets": [
    "aws-cleanup-demo-rahat",
    "test-bucket-01-rahat",
    "test-bucket-02-rahat"
  ]
}
```




Lambda > Functions
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Functions (3)
Last fetched 6/7/2026, 8:02:11 PM
Actions
Create function
Search by attributes or search by keyword
Function name
Description
Package type
Runtime
Type
Last modified
S3-Cleanup
Zip
Python 3.12
Standard
18 minutes ago
Lambda-EC2-Manager
Zip
Python 3.12
Standard
52 minutes ago
DetectUnencryptedS3Buckets
Zip
Python 3.12
Standard
18 seconds ago
Info
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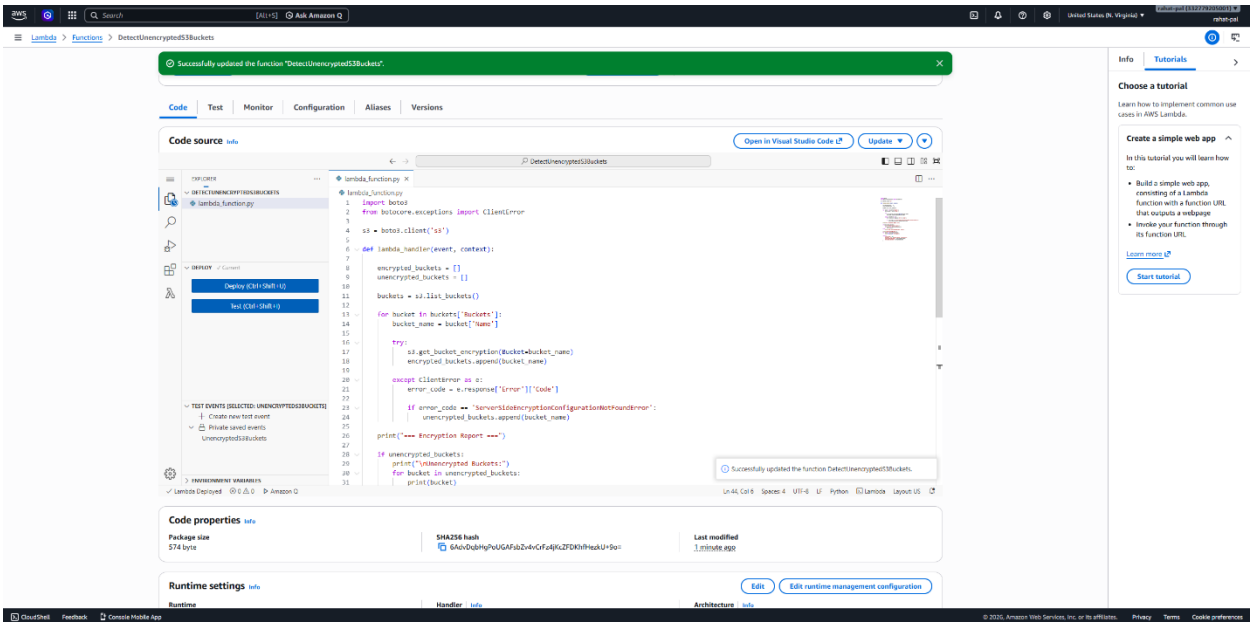
Lambda > Functions > DetectUnencryptedS3Buckets
Successfully updated the function "DetectUnencryptedS3Buckets".
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Update
lambda_function.py
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
Code properties
Package size
574 byte
SHA256 hash
Last modified
1 minute ago
Runtime settings
Edit
Edit runtime management configuration
Runtime
Handler
Architecture
Info
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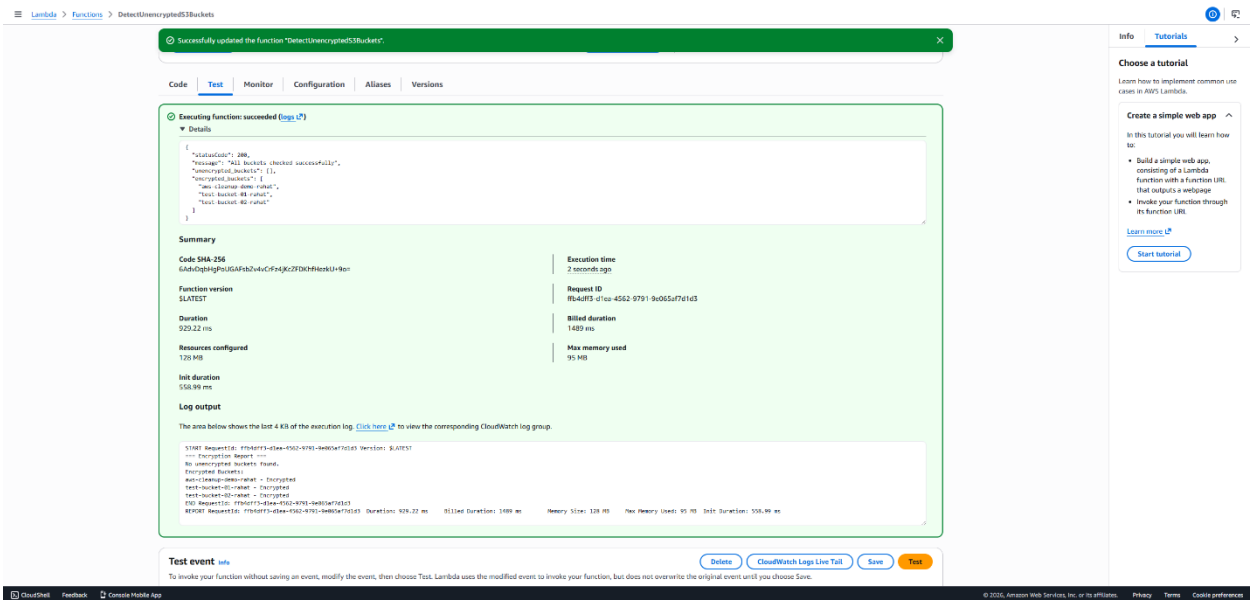
Lambda > Functions > DetectUnencryptedS3Buckets
Successfully updated the function "DetectUnencryptedS3Buckets".
Code
Test
Monitor
Configuration
Aliases
Versions
Executing function: succeeded (logs)
Details
{
"statusCode": 200,
"message": "All buckets checked successfully",
"unencrypted_buckets": [],
"encrypted_buckets": [
"aws-cleanup-demo-rahat",
"test-bucket-01-rahat",
"test-bucket-02-rahat"
]
}
Summary
Code SHA-256
6AdvDqbHgPsUGAFsbZv4vCrFn4jKc2FDKfHHenkU+9on=
Execution time
2 seconds ago
Function version
$LATEST
Request ID
ffb4dff3-d1ea-4562-9791-9e065af7d1d3
Duration
929.22 ms
Billed duration
1489 ms
Resources configured
128 MB
Max memory used
95 MB
Init duration
558.99 ms
Log output
The area below shows the last 4 KB of the execution log. Click here to view the corresponding CloudWatch log group.
START RequestId: ffb4dff3-d1ea-4562-9791-9e065af7d1d3 Version: $LATEST
--- Encryption Report ---
No unencrypted buckets found.
Encrypted Buckets:
aws-cleanup-demo-rahat - Encrypted
test-bucket-01-rahat - Encrypted
test-bucket-02-rahat - Encrypted
END RequestId: ffb4dff3-d1ea-4562-9791-9e065af7d1d3
REPORT RequestId: ffb4dff3-d1ea-4562-9791-9e065af7d1d3 Duration: 929.22 ms Billed Duration: 1489 ms Memory Size: 128 MB Max Memory Used: 95 MB Init Duration: 558.99 ms
Test event
Info
To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.
Delete
CloudWatch Logs Live Tail
Save
Test
Info
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Assignment 4: EBS Snapshot Automation

Step 1: Get Volume ID

Example:

vol-0abcd123456789

Step 2: IAM Role

Attach:

AmazonEC2FullAccess

Step 3: Lambda Code

```
import boto3
from datetime import datetime, timezone, timedelta

ec2 = boto3.client('ec2')

VOLUME_ID = "vol-0abcd123456789"

def lambda_handler(event, context):

   snapshot = ec2.create_snapshot(
      VolumeId=VOLUME_ID,
      Description="Automated Backup"
   )

   print("Created Snapshot:",
        snapshot['SnapshotId'])

   cutoff = datetime.now(timezone.utc) - timedelta(days=30)

   snapshots = ec2.describe_snapshots(
      OwnerIds=['self']
   )

   for snap in snapshots['Snapshots']:

      if snap['StartTime'] < cutoff:

         ec2.delete_snapshot(
            SnapshotId=snap['SnapshotId']
```

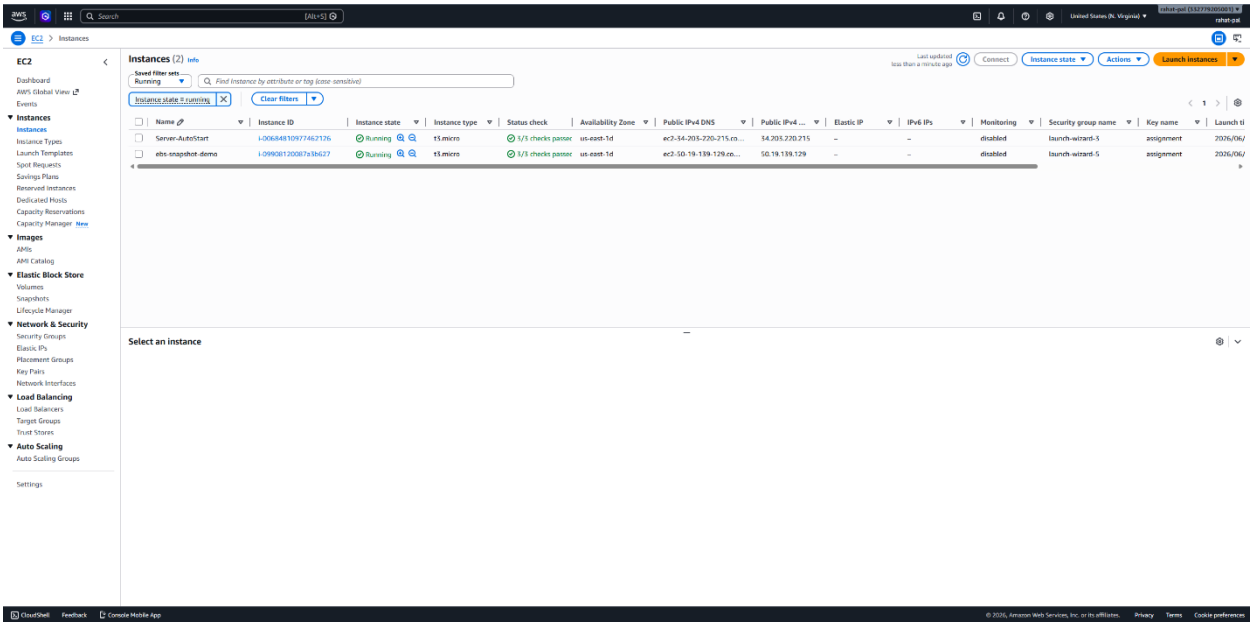
```
        )

        print(
            "Deleted:",
            snap['SnapshotId']
        )

    return {
        'statusCode': 200
    }
```

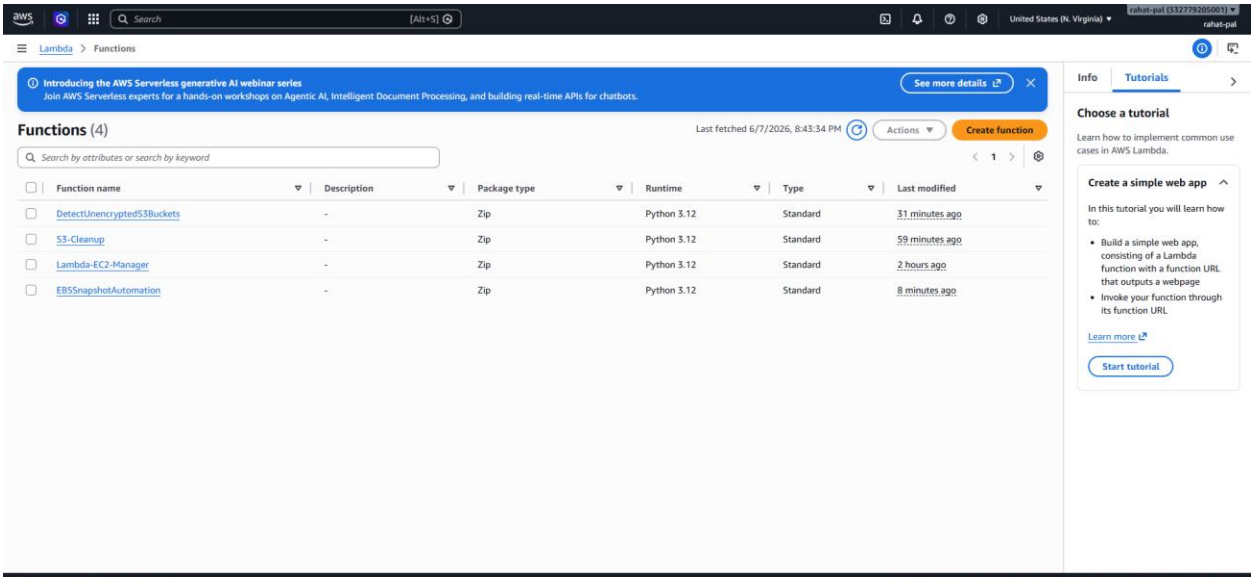
Expected Output

Created Snapshot: snap-01ab234cd567

Deleted: snap-09xy123zz456
Deleted: snap-07gh987jk654

Screenshot

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United States (N. Virginia)
rahat-pal
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Functions (4)
Last fetched 6/7/2026, 8:43:34 PM
Actions
Create function
Search by attributes or search by keyword
Function name
Description
Package type
Runtime
Type
Last modified
DetectUnencryptedS3Buckets - Zip Python 3.12 Standard 31 minutes ago
S3-Cleanup - Zip Python 3.12 Standard 59 minutes ago
Lambda-EC2-Manager - Zip Python 3.12 Standard 2 hours ago
EBSSnapshotAutomation - Zip Python 3.12 Standard 8 minutes ago
Info
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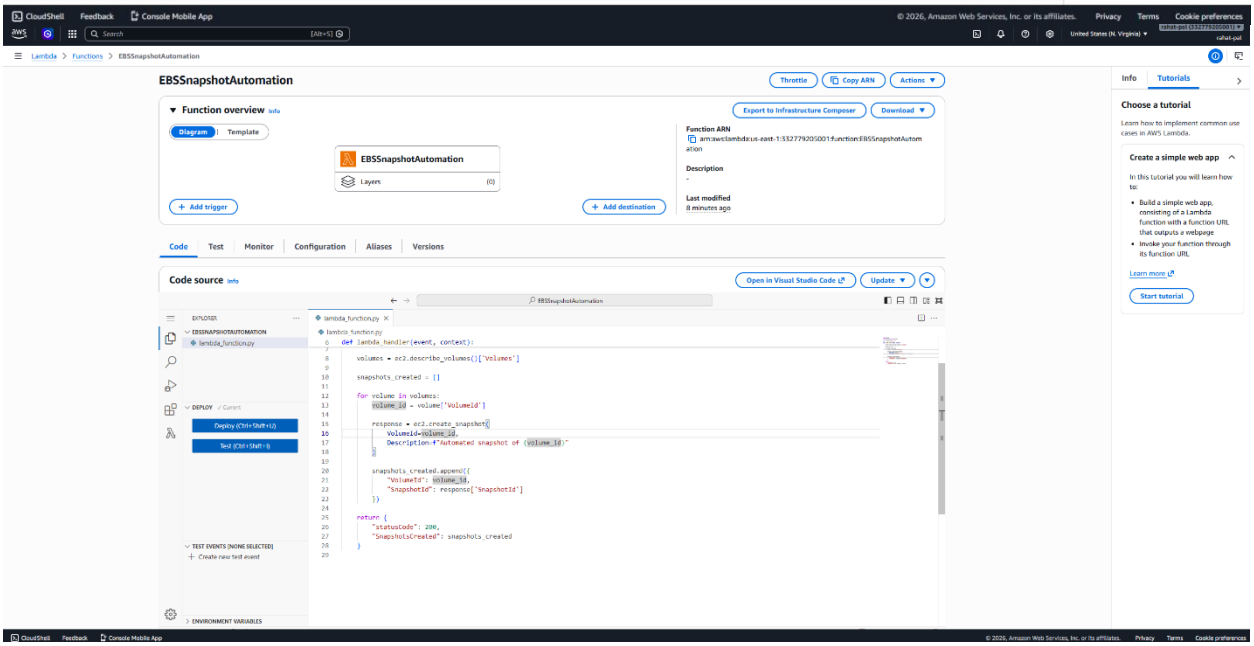
CloudShell
Feedback
Console Mobile App
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Lambda > Functions > EBSSnapshotAutomation
EBSSnapshotAutomation
Throttle
Copy ARN
Actions
Function overview
Export to Infrastructure Composer
Download
Diagram
Template
EBSSnapshotAutomation
Layers
(0)
Add trigger
Add destination
Function ARN
Description
Last modified
8 minutes ago
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Update
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS (NONE SELECTED)
Create new test event
ENVIRONMENT VARIABLES
def lambda_handler(event, context):
volumes = ec2.describe_volumes()['Volumes']
snapshots_created = []
for volume in volumes:
volume_id = volume['VolumeId']
response = ec2.create_snapshot(
VolumeId=volume_id,
Description=f"Automated snapshot of {volume_id}"
)
snapshots_created.append({
"VolumeId": volume_id,
"SnapshotId": response['SnapshotId']
})
return {
"statusCode": 200,
"SnapshotsCreated": snapshots_created
}
Info
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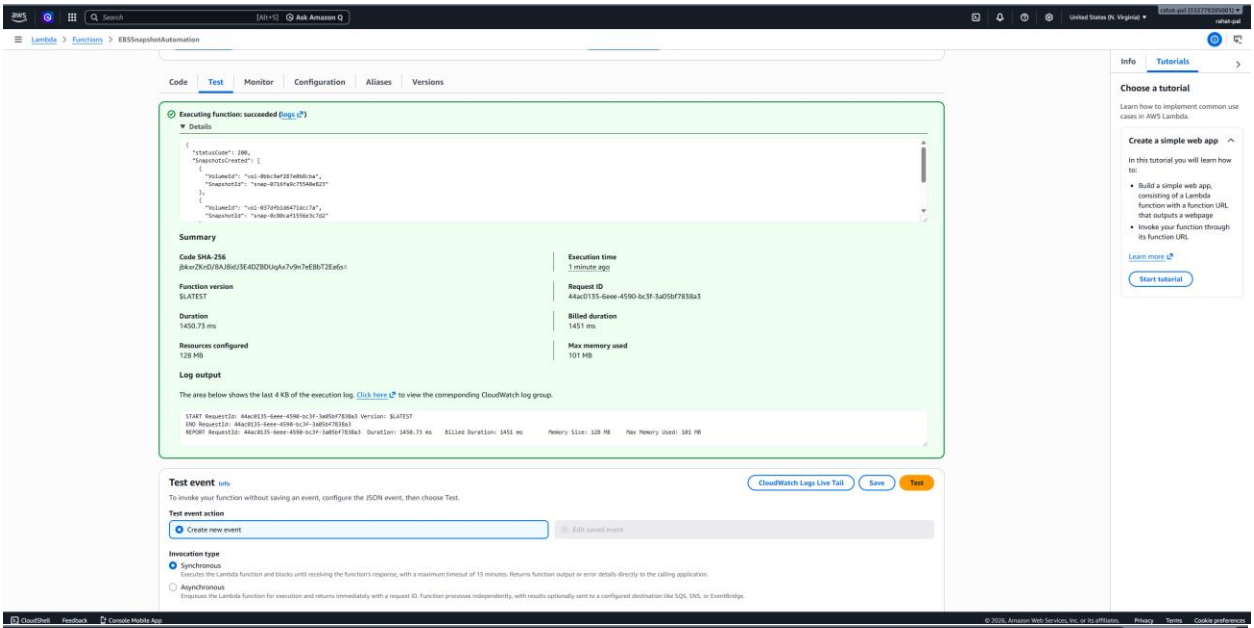
Start tutorial


Lambda > Functions > EBSSnapshotAutomation
Code
Test
Monitor
Configuration
Aliases
Versions
Executing function: succeeded (logs)
Details
Summary
Code SHA-256
Execution time
1 minute ago
Function version
$LATEST
Request ID
44ac0135-6eee-4590-bc3f-3a05bf7838a3
Duration
1450.73 ms
Billed duration
1451 ms
Resources configured
128 MB
Max memory used
101 MB
Log output
The area below shows the last 4 KB of the execution log. Click here to view the corresponding CloudWatch log group.
Test event
Info
To invoke your function without saving an event, configure the JSON event, then choose Test.
CloudWatch Logs Live Tail
Save
Test
Test event action
Create new event
Edit saved event
Invocation type
Synchronous
Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.
Asynchronous
Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.
Info
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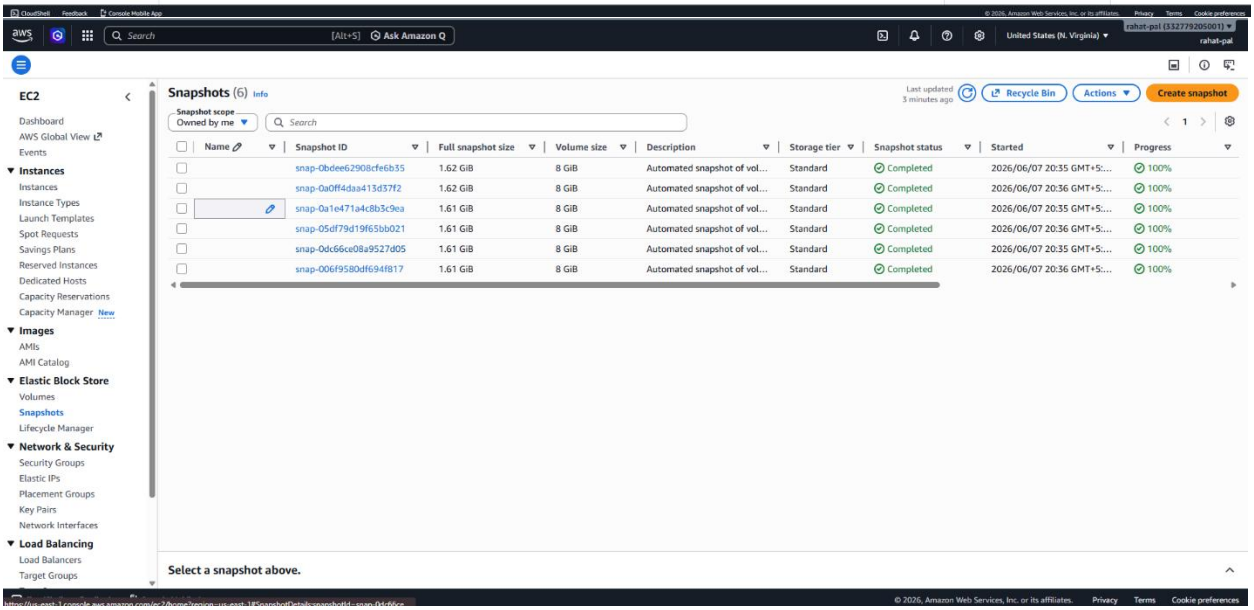
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Snapshots (6) Info
Last updated 3 minutes ago
Recycle Bin
Actions
Create snapshot
Snapshot scope
Owned by me
Search
Name
Snapshot ID
Full snapshot size
Volume size
Description
Storage tier
Snapshot status
Started
Progress
snap-0bdee62908cfe6b55 1.62 GiB 8 GiB Automated snapshot of vol... Standard Completed 2026/06/07 20:35 GMT+5:... 100%
snap-0a0ff4daa413d37f2 1.62 GiB 8 GiB Automated snapshot of vol... Standard Completed 2026/06/07 20:36 GMT+5:... 100%
snap-0a1e471a4c8b3c9ea 1.61 GiB 8 GiB Automated snapshot of vol... Standard Completed 2026/06/07 20:35 GMT+5:... 100%
snap-05df79d19f65bb021 1.61 GiB 8 GiB Automated snapshot of vol... Standard Completed 2026/06/07 20:36 GMT+5:... 100%
snap-0dc66ce08a9527d05 1.61 GiB 8 GiB Automated snapshot of vol... Standard Completed 2026/06/07 20:35 GMT+5:... 100%
snap-006f9580df694f817 1.61 GiB 8 GiB Automated snapshot of vol... Standard Completed 2026/06/07 20:36 GMT+5:... 100%
Select a snapshot above.